

MLOps CI/CD Experiment Report

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Objective

To implement a basic MLOps CI/CD pipeline using GitHub Actions for automated testing and deployment packaging of a machine learning project.

Brief Description of the MLOps CI/CD Pipeline

The project uses an Iris dataset and a Logistic Regression machine learning model. The model code is tested using pytest. GitHub Actions automatically runs the tests whenever changes are pushed to the main branch. After the testing stage succeeds, the Continuous Delivery stage creates a deployment package containing the model, requirements, and tests, and uploads it as a GitHub Actions artifact.

Experiment Result

The machine learning model was successfully tested using pytest. The experiment produced the result: 1 test passed. The GitHub Actions workflow successfully performs the CI testing stage and the CD packaging stage.

Conclusion

The experiment successfully demonstrated an MLOps CI/CD workflow. The project automates model testing through Continuous Integration and creates a deployment package through Continuous Delivery. This improves the reliability and repeatability of the machine learning workflow.